Process of the Chi-Square Test

The Chi-Square test follows a systematic procedure to determine whether the observed differences between categorical variables are statistically significant.

Step 1: Define the Research Problem

Clearly identify the research question and determine whether the variables involved are categorical.

Example

"Is there a relationship between gender and online shopping preference?"

Step 2: Formulate the Hypotheses

Develop the null and alternative hypotheses.

Null Hypothesis (H_0):

There is **no significant association** between gender and online shopping preference.

Alternative Hypothesis (H_1):

There **is a significant association** between gender and online shopping preference.

Step 3: Select the Significance Level

Choose the level of significance (**α**).

Usually:

- $\alpha = 0.05$ (95% confidence)
- $\alpha = 0.01$ (99% confidence)

Step 4: Collect and Organize Data

Collect data and prepare a contingency table.

Gender	Prefer Online	Prefer Offline	Total
Male	40	20	**60**
Female	35	25	**60**
Total	**75**	**45**	**120**

Step 5: Calculate Expected Frequencies

Expected frequency for each cell is calculated using:

$$E = \frac{(\text{Row Total})(\text{Column Total})}{\text{Grand Total}}$$

For Male–Online:

$$E = \frac{60 \times 75}{120} = 37.5$$

Repeat for all cells.

Step 6: Calculate the Chi-Square Statistic

Use the formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Calculate each cell's contribution and sum them to obtain the calculated Chi-Square value.

Step 7: Determine Degrees of Freedom

Degrees of Freedom (df):

$$df = (r - 1)(c - 1)$$

Where:

- **r** = Number of rows
- **c** = Number of columns

For a 2 × 2 table:

$$df = (2 - 1)(2 - 1) = 1$$

Step 8: Find the Critical Value or p-value

Using:

- Chi-Square Distribution Table, or
- Statistical software (SPSS, R, Excel)

At $\alpha = 0.05$ and $df = 1$:

Critical χ^2 = **3.841**

Chi-square Distribution Table

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57
22	8.64	9.54	10.98	12.34	14.04	30.81	33.92	36.78	40.29
24	9.89	10.86	12.40	13.85	15.66	33.20	36.42	39.36	42.98
26	11.16	12.20	13.84	15.38	17.29	35.56	38.89	41.92	45.64
28	12.46	13.56	15.31	16.93	18.94	37.92	41.34	44.46	48.28
30	13.79	14.95	16.79	18.49	20.60	40.26	43.77	46.98	50.89
32	15.13	16.36	18.29	20.07	22.27	42.58	46.19	49.48	53.49
34	16.50	17.79	19.81	21.66	23.95	44.90	48.60	51.97	56.06
38	19.29	20.69	22.88	24.88	27.34	49.51	53.38	56.90	61.16
42	22.14	23.65	26.00	28.14	30.77	54.09	58.12	61.78	66.21
46	25.04	26.66	29.16	31.44	34.22	58.64	62.83	66.62	71.20
50	27.99	29.71	32.36	34.76	37.69	63.17	67.50	71.42	76.15
55	31.73	33.57	36.40	38.96	42.06	68.80	73.31	77.38	82.29
60	35.53	37.48	40.48	43.19	46.46	74.40	79.08	83.30	88.38
65	39.38	41.44	44.60	47.45	50.88	79.97	84.82	89.18	94.42
70	43.28	45.44	48.76	51.74	55.33	85.53	90.53	95.02	100.43
75	47.21	49.48	52.94	56.05	59.79	91.06	96.22	100.84	106.39
80	51.17	53.54	57.15	60.39	64.28	96.58	101.88	106.63	112.33
85	55.17	57.63	61.39	64.75	68.78	102.08	107.52	112.39	118.24
90	59.20	61.75	65.65	69.13	73.29	107.57	113.15	118.14	124.12
95	63.25	65.90	69.92	73.52	77.82	113.04	118.75	123.86	129.97
100	67.33	70.06	74.22	77.93	82.36	118.50	124.34	129.56	135.81

Step 9: Make the Decision

Decision Rule:

- If **χ^2 calculated** > **χ^2 critical**, reject H_0.
- If **χ^2 calculated** ≤ **χ^2 critical**, fail to reject H_0.

Alternatively:

- If **$p \leq 0.05$**, reject H_0.
- If **$p > 0.05$**, fail to reject H_0.

Step 10: Interpret the Results

Interpret the findings in the context of the research problem.

Working Example

Research Problem

A researcher wants to determine whether **gender influences online shopping preference**.

Step 1: Hypotheses

H_0: There is no association between gender and online shopping preference.

H_1: There is an association between gender and online shopping preference.

Step 2: Observed Frequencies

Gender	Online	Offline	Total
Male	40	20	60
Female	35	25	60
Total	**75**	**45**	**120**

Step 3: Expected Frequencies

Gender	Online	Offline
Male	37.5	22.5
Female	37.5	22.5

Step 4: Calculate χ^2

Using the formula:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

The calculated value is:

$\chi^2 = 0.89$

Step 5: Degrees of Freedom

$$df = (2 - 1)(2 - 1) = 1$$

Step 6: Critical Value

At $\alpha = 0.05$ and df = 1:

Critical $\chi^2 = 3.841$

Step 7: Decision

Since:

$$0.89 < 3.841$$

Fail to reject the null hypothesis.

Step 8: Interpretation

There is **no statistically significant association** between gender and online shopping preference. The observed differences may have occurred due to chance.

Reporting the Result

A Chi-Square test of independence was conducted to examine the relationship between gender and online shopping preference. The results indicated that the association was not statistically significant, $\chi^2(1) = 0.89$, $p > 0.05$*. Therefore, the null hypothesis was not rejected, suggesting that gender and online shopping preference are independent.*